

Technical Documentation - RAG Customer Support System

1. Introduction

RAG combines retrieval + generation to improve factual accuracy in LLM systems.
Used for customer support chatbot over PDF knowledge base.

2. System Architecture

Includes ingestion pipeline, vector DB, retriever, LLM, LangGraph orchestration, and HITL layer.

3. Design Decisions

Chunk size: 800 tokens for balance.
Overlap: 200 tokens.
ChromaDB for simplicity and speed.
Prompt engineered for grounding responses.

4. Workflow

LangGraph controls flow between retrieval and generation nodes using state transitions.

5. Conditional Logic

Routes queries based on context availability and confidence scores.

6. HITL Implementation

Human intervention ensures correctness in uncertain queries.

7. Challenges

Balancing accuracy vs latency.
Chunk optimization.
Context relevance issues.

8. Testing

Tested with FAQ-style queries and ambiguous inputs.

9. Future Enhancements

Multi-document support.
FastAPI deployment.
Feedback learning loop.
Persistent memory.